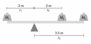
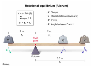


The figure above shows a board placed on top of a fulcrum that is balanced by three masses (M_1, M_2, and M_3). If the mass of M_1 and M_2 are 60 kg and 30 kg, respectively, what is the mass of M_3? (Note: Assume that the weight of the board is negligible.)

- A. 8.5 kg [6%]
- ✔ B. 17 kg [74%]
- C. 34 kg [15%]
- D. 68 kg [3%]

Correct
75% answered correctly

Explanation:



Torque is the tendency of an **applied force** to **cause the rotation** of an object around a **pivot point** located at some **radial distance** from where the force is applied. The magnitude of torque (τ) is expressed in terms of the magnitude of the applied force (F), the radial distance (r), and the angle (θ) between F and r:

$$\tau = r \cdot F \sin(\theta)$$

Because torque increases as a force of equal magnitude is applied at greater distances from the pivot point, a mechanical advantage is observed in the context of rotational forces applied to a lever arm. However, when the **sum of all torques** acting on an object is **equal to zero**, an object is in **rotational equilibrium** (no rotation occurs):

$$\sum_{\text{Torques}} = 0$$

In this question, a board placed on a fulcrum is balanced by three separate masses. Each mass exerts a force on the board due to its weight. The resulting torque exerted on the left side of the fulcrum by the weight of M_1 is counterbalanced by the two torques exerted on the right side of the fulcrum by the weights of M_2 and M_3:

$$\sum_{\text{left}}(\text{torques}) = \sum_{\text{right}}(\text{torques})$$

Substituting the expressions for torque into the summation equation gives:

$$F_1 \cdot r_1 \sin(\theta_1) = F_2 \cdot r_2 \sin(\theta_2) +$$

Representing each torque as a product of weight and radial distance from the fulcrum yields:

$$M_1 \cdot g \cdot r_1 \sin(\theta_1) = [M_2$$

Because the acceleration due to gravity g cancels from all terms and because all angles are equal to 90° (sin 90° = 1), the expression simplifies to:

$$M_1 \cdot \cancel{g} \cdot r_1 (1) = [M_2 \cdot \cancel{g} \cdot r_2$$

$$M_1 \cdot r_1 = M_2 \cdot r_2 + M_3 \cdot r_3$$

Using the values of M_1, M_2, r_1, and r_2 provided in the question yields:

$$60\ \text{kg} \cdot 2\ \text{m} = (30\ \text{kg} \cdot 2\ \text{m}) +$$

$$120\ \text{kg} \cdot m = (60\ \text{kg} \cdot \text{m}) + (M_3 \cdot 3.5\ \text{m})$$

$$60\ \text{kg} \cdot \text{m} = M_3 \cdot 3.5\ \text{m}$$

$$M_3 \approx 17\ \text{kg}$$

Therefore, the mass of M_3 must be approximately 17 kg for the board to remain in rotational equilibrium **(Choice B)**.

Educational objective:
Torque is the tendency of an applied force to cause an object to rotate around a pivot point located at some radial distance from where the force is applied. Rotational equilibrium occurs when the sum of all torques acting on an object is equal to zero and results in no rotation.

Time spent: 0 sec
Subject: Physics

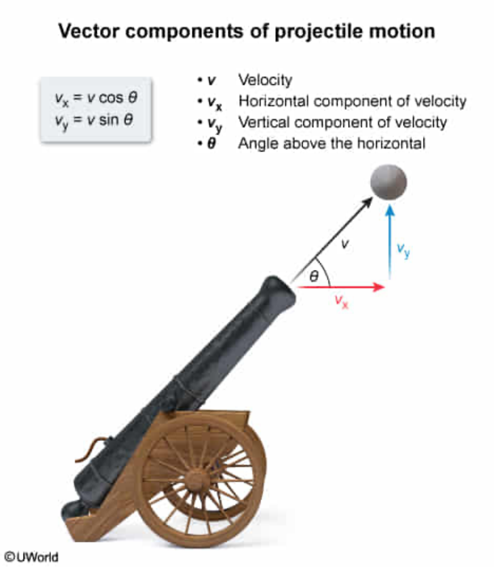
QID: 402546
System: Mechanics and Energy

A cannonball is launched at a speed of 30 m/s and at an angle of 60° above the horizontal. What are the magnitudes of the horizontal and vertical components of the cannonball's initial velocity, respectively?

- A. 5 m/s and 25 m/s [3%]
- B. 10 m/s and 15 m/s [8%]
- C. 15 m/s and 25 m/s [72%]
- D. 25 m/s and 15 m/s [15%]

Correct
72% answered correctly

Explanation:



Unlike scalar quantities, **vectors** contain both a **magnitude** and a **direction**. Vectors may be added to or subtracted from other vectors. Consequently, a vector located within a two-dimensional coordinate system can be represented as two vector components in which one vector component is oriented along the *x*-axis and the other is oriented along the *y*-axis. Mathematically, the overall (resultant) vector is the hypotenuse (longest side) of a right triangle in which the *x*- and *y*-components of the vector form the adjacent and opposite sides.

Furthermore, the angle (θ) between the overall vector and its adjacent, horizontal (*x*-axis) component can be used with trigonometric ratios to calculate the magnitude of the horizontal component (V_x) and vertical component (V_y) of the initial vector (V):

$$V_x = V \cdot \cos\ \theta$$

$$V_y = V \cdot \sin\ \theta$$

In this question, the cannonball is launched with an initial speed of 30 m/s from a cannon angled 60° above the horizontal (ie, the ground). When a convenient orientation is selected for an *xy*-coordinate

system in which the ground is parallel to the *x*-axis, the cannonball's velocity vector (V) vector is oriented 60° above the *x*-axis with a magnitude (length) of 30 m/s. Accordingly, the magnitudes of the horizontal (V_x) and vertical components (V_y) of velocity can be calculated as:

$$v_x = 30\ \frac{\text{m}}{\text{s}} \cdot \cos(60°) v_y = 30\ \frac{\text{m}}{\text{s}} \cdot \sin(60°)$$

$$v_x = 30\ \frac{\text{m}}{\text{s}} \cdot \frac{1}{2} = 15\ \frac{\text{m}}{\text{s}} v_y = 30\ \frac{\text{m}}{\text{s}} \cdot \frac{\sqrt{3}}{2} \approx 25\ \frac{\text{m}}{\text{s}}$$

Therefore, the horizontal component of the cannonball's initial velocity has a magnitude of 15 m/s, and the vertical component of the cannonball's initial velocity has a magnitude of 25 m/s **(Choice C)**.

Educational objective:
Vector quantities can be separated into fundamental component vectors using trigonometric ratios. The horizontal and vertical components of an object's velocity can be calculated using the magnitude of the velocity and its angle above the horizontal.

Time spent:0 sec
Subject:
Physics

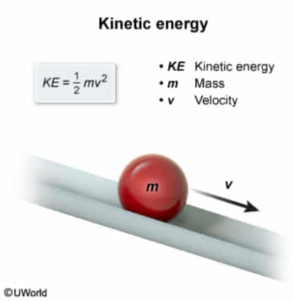
QID:402547
System:
Mechanics and Energy

A marble rolls down a slope at a velocity of 10 m/s. If the kinetic energy of the marble is 2 J, what is the mass of the marble?

- A. 20 g [12%]
- B. 40 g [84%]
- C. 60 g [1%]
- D. 80 g

Correct
84% answered correctly

Explanation:



Gravitational potential energy and kinetic energy are two forms of energy that an object may possess in mechanical systems. Gravitational potential energy is energy that is stored based on an object's position relative to a reference point within a gravitational field. Consequently, potential energy is often referred to as the "energy of position."

Kinetic energy is the energy possessed by an object in motion, whether that object is rotating or moving through space via translational motion. Consequently, kinetic energy may be referred to as the "energy of motion," and the magnitude of the kinetic energy (*KE*) is determined by the mass (*m*) and velocity (*v*) of a moving object:

$$KE = \frac{1}{2} m \cdot v^2$$

The SI unit for energy is the joule (J), where 1 joule is equal to the product of mass (kg) and velocity (m/s) squared (1 J = 1 $kg{\cdot}m^2{\cdot}s^{-2}$).

Certain mechanical processes may convert energy from one form to another. For example, an object falling from a height loses gravitational potential energy and gains kinetic energy. However, the total energy of an object is always conserved.

In this question, a marble has 2 J of kinetic energy as it rolls down a slope at a velocity of 10 m/s. Provided that no forces other than gravity act on the marble to cause motion, it can be assumed that all of this kinetic energy was attained through the conversion of its gravitational potential energy.

The mass of the marble can be determined by substituting given values and solving for mass in the equation for the kinetic energy of an object:

$$2\ \mathrm{J} = \frac{1}{2} m \cdot \left(10\ \frac{\mathrm{m}}{\mathrm{s}}\right)^2$$

$$m = \frac{4\ \mathrm{J}}{100\ \frac{\mathrm{m}^2}{\mathrm{s}^2}}$$

$$m = \frac{4\ \mathrm{kg} \cdot \cancel{\frac{\mathrm{m}^2}{\mathrm{s}^2}}}{100\ \cancel{\frac{\mathrm{m}^2}{\mathrm{s}^2}}} = 0.04$$

Therefore, the marble has a mass of 0.04 kg, or 40 g **(Choice B)**.

Educational objective:
Kinetic energy (often expressed in SI units using the joule) is a form of mechanical energy that has a magnitude directly proportional to the mass and the square of the velocity of an object.

Time spent: 0 sec
Subject: Physics

QID: 402548
System: Mechanics and Energy